

Model Card Evaluation Report

1. Model Details

Developer: Machine Learning Specialist Candidate

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Model Architecture: Logistic Regression Classifier using the liblinear optimization solver.

Target Objective: Predictive classification of individual annual income, where a positive classification (1) represents income exceeding \$50,000 annually, and a negative classification (0) represents income less than or equal to \$50,000.

2. Intended Use

Primary Scope: This model was developed exclusively for educational benchmarking, systemic algorithmic auditing, and fairness mitigation research.

Prohibited Scopes: This system is explicitly not designed or authorized for production-level automated recruitment, human resource screening, individual credit rating generation, or loan underwriting systems.

3. Factors and Demographics

Monitored Cohorts: Stratified analysis was conducted on biological sex demographics.

Protected Attribute: Gender representation serves as the primary operational factor, mapped internally to binary values where male data tracks as the historically privileged cohort and female data tracks as the unprivileged cohort.

4. Operational Metrics

Performance Evaluation: Group-specific classification accuracy and the Area Under the Receiver Operating Characteristic Curve (ROC AUC).

Fairness Standard: Disparate Impact Ratio evaluation, applying the standard regulatory four-fifths rule threshold of 0.80 to determine systemic classification disparity.

5. Training and Evaluation Datasets

Data Origin: Cleaned extractions from the official UCI Adult Income dataset sourced dynamically via OpenML.

Data Splits: Standard random partition consisting of an 80% subset allocated for training and a 20% subset reserved for evaluation testing.

Predictive Features: Chronological age, sequential education numeric rank, and the biological sex indicator.

6. Quantitative Analysis and Bias Mitigation

Initial Evaluation: The baseline disparate impact metric registered significantly below the 0.80 threshold, confirming substantial negative classification skew regarding female records inherited from historical datasets.

Mitigated Evaluation: Post-application of the IBM AI Fairness 360 preprocessing reweighing algorithm, sample group weights adjusted dynamically to yield a corrected disparate impact metric of approximately 1.0000.

Performance Margin: Systemic equity corrections preserved baseline classification accuracy bounds within a negligible performance variance margin of less than 1.5%.

7. Explanations and Interpretability

Global Interpretability: SHAP linear feature attributions isolate education level as the primary structural vector driving positive income classifications.

Feature Trajectories: Increasing values for educational duration and chronological age show direct, linear transitions into positive SHAP value ranges, directly raising model predictive confidence toward high-income classification thresholds.

8. Ethical Risks and Technical Limitations

Historical Reflection: The underlying census distributions capture structural wage anomalies from past decades, meaning unmitigated runs simply automate historical societal imbalances.

Proxy Risks: Apparent neutral criteria such as formal education length can inadvertently mimic broader socioeconomic or localized structural stratification.

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